that each of the five string formulations involves two-dimensional membranes as well.

This raises two questions. First, are two-dimensional membranes the true fundamental ingredient of string theory? And second, having made the bold leap in the 1970s and early 1980s from zero-dimensional point particles to one-dimensional strings, and having now seen that string theory actually involves two-dimensional membranes, might it be that there are even higher-dimensional ingredients in the theory as well? As of this writing, the answers to these questions are not fully known, but the situation appears to be the following.

We relied heavily on supersymmetry to give us some understanding of each formulation of string theory beyond the domain of validity of perturbative approximation methods. In particular, the properties of BPS states, their masses and their force charges, are uniquely determined by supersymmetry, and this allowed us to understand some of their strongly coupled characteristics without having to perform direct calculations of unimaginable difficulty. In fact, through the initial efforts of Horowitz and Strominger, and through subsequent groundbreaking work of Polchinski, we now know even more about these BPS states. In particular, not only do we know their masses and the force charges they carry, but we also have a clear understanding of what they look like. And the picture is, perhaps, the most surprising development of all. Some of the BPS states are one-dimensional strings. Others are two-dimensional membranes. By now, these shapes are familiar. But, the surprise is that yet others are three-dimensional, four-dimensional—in fact, the range of possibilities encompasses every spatial dimension up to and including nine. String theory or M-theory, or whatever it is finally called, actually contains extended objects of a whole slew of different spatial dimensions. Physicists have coined the term three-brane to describe extended objects with three spatial dimensions, four-brane for those with four spatial dimensions, and so on up to nine-branes (and, more generally, for an object with p space dimensions, where p represents a whole number, physicists have coined the far from euphonious terminology p-brane). Sometimes, using this terminology, strings are described as one-branes, and membranes as two-branes. The fact that all of these